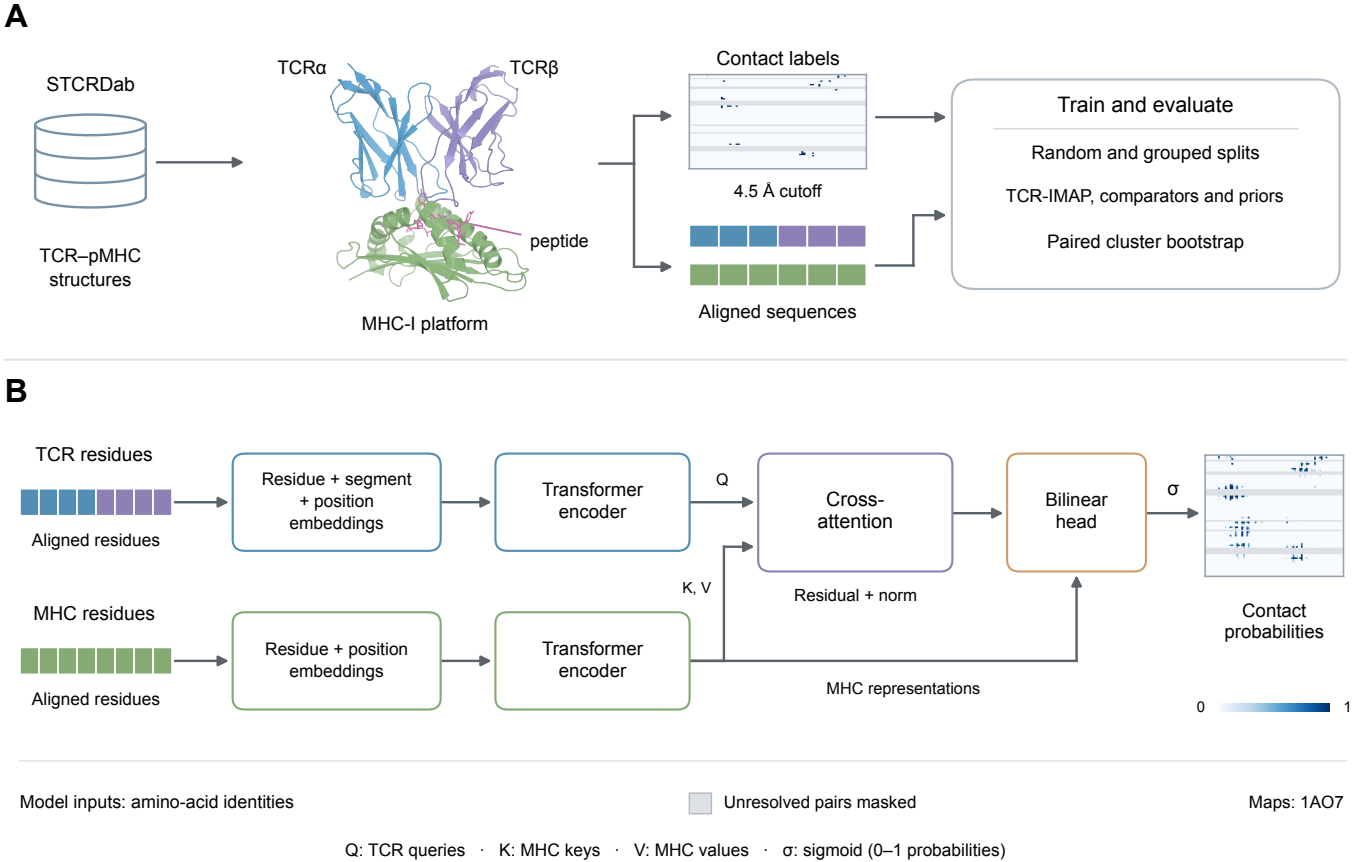

A
STCRDab
TCR–pMHC structures
TCRα
TCRβ
peptide
MHC-I platform
Contact labels
4.5 Å cutoff
Aligned sequences
Train and evaluate
Random and grouped splits
TCR-IMAP, comparators and priors
Paired cluster bootstrap
B
TCR residues
Aligned residues
Residue + segment + position embeddings
Transformer encoder
Q
Cross-attention
Residual + norm
Bilinear head
σ
Contact probabilities
0
1
K, V
MHC residues
Aligned residues
Residue + position embeddings
Transformer encoder
MHC representations
Model inputs: amino-acid identities
Unresolved pairs masked
Maps: 1AO7
Q: TCR queries · K: MHC keys · V: MHC values · σ: sigmoid (0–1 probabilities)